



AUTO REMEDIATION TOOL



Contents

Architecture and Components	2
Overview	2
Architecture.....	2
Implementation.....	3
NMS Alerts and Thresholds	3
Data lake and Sync	4
Analyze Module APIs and Logic	5
Alert API:.....	5
Alert Soaking	5
Auto Remediation Flow:.....	5
Understanding Transaction Status	7
Reverse Remediation (Applicable only for Cases where Cost Change is Done)	7
Details of Shutdown and Cost Change Operations.....	8
Accessing the Tool and Debugging	9
Work Chat Alerts	12

Auto Remediation Tool

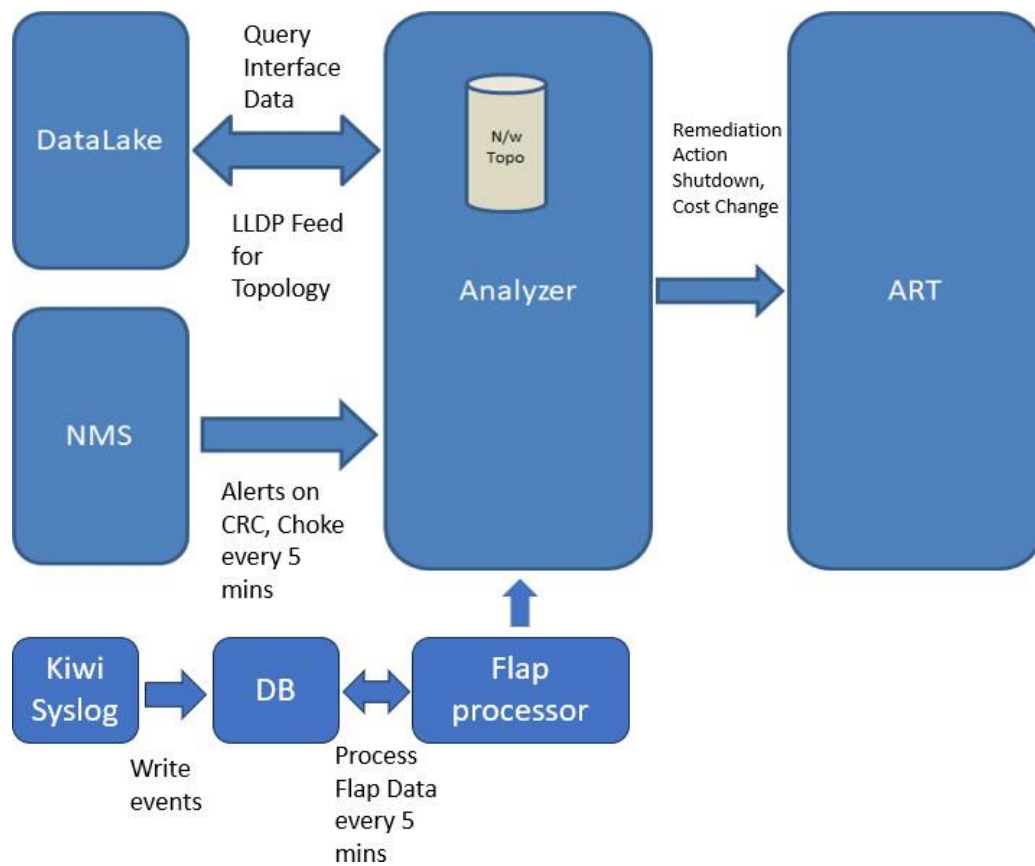
Architecture and Components

Overview

The Auto Remediation System Detects Issues in the Network, Identifies the Segment Impacted and Finds any Alternate path (if available) where the traffic can be re-routed. It then Triggers the change in the network based on the paths Identified.

The Scope of this tool is remediation of CRC, Choking and Port Flap events in the Access network (DX-AX)

Architecture



Following are the Key Components in the System:

- **Data Lake:** The Data lake acts as a repository for the entire Network Data. LLDP, Bundle Data, Network Monitoring Data (Sourced from NMS).
- **NMS:** NMS is a key component where the Alerting Logic and Thresholds are configured. Based on the conditions, NMS sends an event to the Analyzer Module for action.
- **N/W Topology (neo4J DB):** A Graph Database which stores the physical topology information of the network.
- **ART:** This tool performs all the device related operations for Remediation. Currently supports link shutdown and cost change operations.
- **Analyzer:** This is the brain of the entire system. It builds the topology Database from Data lake. It Receives, Soaks the alerts received from NMS. It is also responsible to identify the alternate Paths, validate the paths and then form a trigger to ART for appropriate Action.

Implementation

NMS Alerts and Thresholds

All the Alert Configurations are done in the Solar Winds NMS System.

The Alerts are configured for Following N/W Anomalies:

1. **CRC Errors:**

The Alert is configured based on the Interface In Packets and Interface in errors. The data is polled every 5 Minutes and is evaluated every 5 minutes.

Threshold Calculation:

$\text{Error\%} = (\text{interface in errors} / \text{interface in packets}) \times 100$

If $\text{Error\%} \geq 0.01$ then the Alert is active

If $\text{Error\%} < 0.01$ then Alert is Cleared

Alerting Mechanism:

Every 5 mins the condition is checked. If the Alert condition for active alert is satisfied, then an Alert is triggered to Analyze system.

Once an Alert is active, every 5 minutes the active alert event is sent to Analyze System until the condition is satisfied.

Once the Active Condition clears, then a single Clear Alert is sent to Analyze System.

2. **Choking:**

The alert is Configured based on the bandwidth utilisation polled by the NMS. The data is polled every 5 Minutes and is evaluated every 5 minutes.

Threshold Calculation:

Bandwidth $\geq 95\%$ then the Alert is active

Bandwidth < 95% then the Alert is cleared.

Alerting Mechanism:

Every 5 mins the condition is checked. If the Alert condition for active alert is satisfied, then an Alert is triggered to Analyze system.

Once an Alert is active, every 5 minutes the active alert event is sent to Analyze System until the condition is satisfied.

Once the Active Condition clears, then a single Clear Alert is sent to Analyze System.

3. Link Flap:

The alert is Configured based on the Syslogs received from the Kiwi Syslog Server in each City. These Logs are written into a Database, which is then further Processed by a Python Module (Flaps processor)

Following Logs are a part of the Filter

- "SNMP_TRAP_LINK_DOWN.*et" (Juniper)
- "SNMP_TRAP_LINK_DOWN.*xe" (Juniper)
- "LINKUPDOWN.*Gigabit.*down" (Huawei)
- "IF_STATE\(\).*GigabitEthernet.*DOWN" (Huawei)
- "IF_STATE\(\).*GE.*DOWN" (Huawei)
- "SNMP-WARNING-linkDown.*Interface [0-9]\/[0-9]\/.*[0-9]" (Nokia)

Threshold Calculation:

For two consecutive 5 Minute interval, >=3 Flaps are found then the alert is sent as Link Flapping and ART will perform the Remediation.

Alerting Mechanism:

Every 5 mins the Link Flap events are fetched and counted. If >=3 then the trigger is sent to ART.

If an event is not received for a given link for >=5 mins then ART will auto clear the event in its DB.

Data lake and Sync

The following Key Data points are collected from Data lake:

1. LLDP Master Data: This data is used to populate the NEO4J Database in the Analyze System. This populates Device Details (ID, IP, Name, Segment), interface Connectivity (ID, Name, Neighbour), Bundled Interfaces etc in the NEO4J DB.
The LLDP Data is currently Pulled on a Scheduled Basis once a Day and updated.
2. Node Table: This is used to get the OEM, Model Details of the Devices.
3. Interface Status latest: This table contains the last polled status of the Interfaces. It provides the Interface Status, Speed, CRC, Packets, Traffic.
4. Optic power: This Table has the Optic power data last populated for the interfaces. This is currently refreshed once a day.
5. Interface Flap: This Data contains the Port up, Down events for all the interfaces.
6. Interface Traffic: This contains peak traffic grouped for every Hour. This is used to calculate the peak traffic flowing in the links/bundle.

Analyze Module APIs and Logic

Alert API:

The Alert API is called by NMS to notify of an event. The supported Issues currently are CRC, CHOKe, FLAP.

http://localhost:8001/api/create/transaction?node_name=BLR-KLK-AX1&interface_name=XGigabitEthernet0/1/3&alert_status=DOWN&city=Bangalore&transaction_issue=CRC&segment=AX&datalake_time=324324.32432&total_packets=100000&crc_count=1&bandwidth=15

node_name: is the node for which the alert is generated

interface_name: is the interface that has the issue

alert_status: DOWN if Alert is Active, UP if Alert is Cleared

city: city

transaction_issue: CRC, CHOKe, FLAP based on the issue

segment: The type of the device that the issue occurred on

datalake_time: time of issue

total_packets: This is the total packet count recorded. Used only if ISSUE is CRC.

crc_count: Polled CRC Counter value for last interval. Used only if ISSUE is CRC.

Bandwidth: Interface Bandwidth in %. Used only If ISSUE is CHOKe.

Alert Soaking:

CRC, CHOKe, FLAP:

- Soaking of Alert is Implemented to act on issues that occur over multiple intervals only.
- Soaking interval for CRC Errors is 10 minutes and action is taken only if 2 DOWN Alerts are received in 10 Minutes.
- All other Alerts are ignored.

Auto Remediation Flow:

Upon receiving an Alert that satisfies the Soaking Logic the System creates a transaction and Puts the Transaction in Queue for processing.

Figures 1, 2 and 3 explain the Flow of the queue.

Figure 1

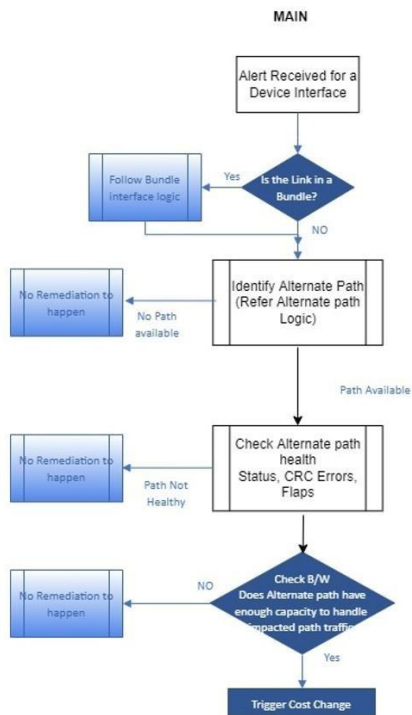


Figure 2

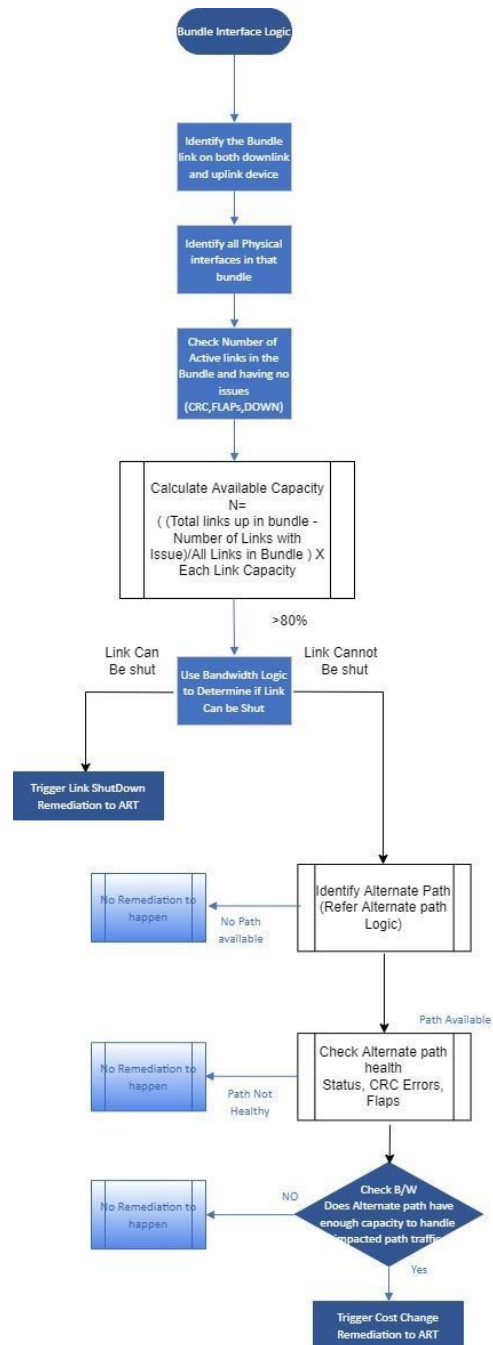
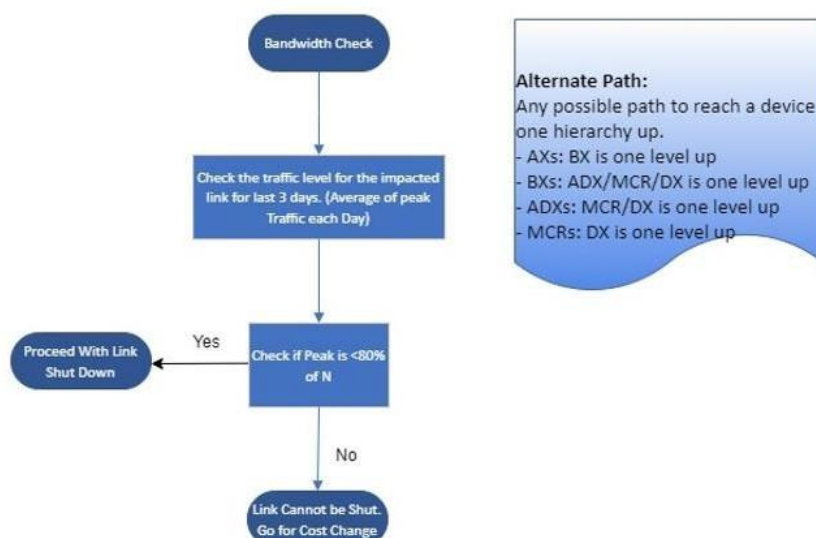


Figure 3



Understanding Transaction Status

S No	TXN State	Details
1	SOAK_REJECT	Alert Rejected as it is the First alert in soaking interval
2	SOAK_ACCEPT	Alert Accepted as it satisfied the Soaking Logic
3	NODE_ERROR	Node or Interface not found in DB
4	IN_QUEUE	Transaction put in queue for processing
5	PATH_COMPUTE	Alternate path being Computed
6	ALTERNATE_PATH_NOT_FOUND	No Alternate paths found or No alternate Paths Healthy
7	ALTERNATE_PATH_NOT_QUALIFIED	A Specific Path did not get qualified
8	PATH_SELECTION	Alternate Paths being evaluated for status etc
9	REMEDIATION_SUCCESFUL	Successfully Remediated
10	REMEDIATION_FAILED	Remediation Failed
11	REVERSE_REMEDIATION_SUCCESFUL	Reverse Remediation for this Transaction is successful
12	HIVEDB_ERROR	There was an error while communicating with the Analytics Database

Reverse Remediation (Applicable only for Cases where Cost Change is Done)

Reverse Remediation is the reversal of Changes done.

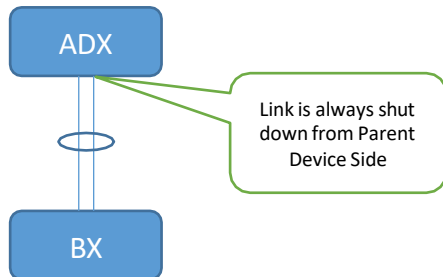
This is scheduled for all Cost Change Remediation that have happened and have crossed 24 Hours.

This runs at 5 AM every day and changes the cost back to the value it was before remediation happened.

Details of Shutdown and Cost Change Operations:

Shutdown:

If the System Determines that a link with issue can be shut down, Always the Downlink of the Parent Device is Shut Down. The tool never Shuts Down the Uplink Interface.



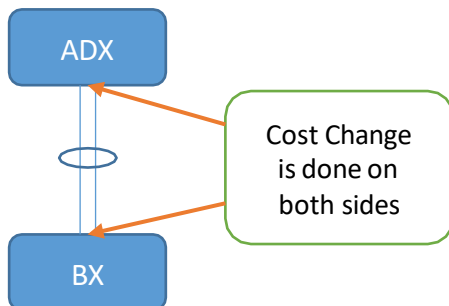
Cost Change:

The Cost Change Operation Increases the cost by a Factor of 500.

If the earlier cost on interface was 10, it will become 510. If it was 50, it will become 550 and so on.

The system also remembers the old Cost to reverse the cost back after 24 Hours.

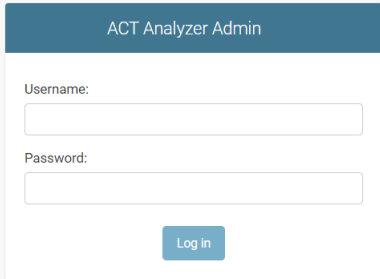
If the Link belongs to a bundle, the cost change is done on the bundle. If the link is an individual Link then the Cost change is done on the interface.



Accessing the Tool and Debugging

1. Logging into the Portal:

Enter the Username and Password to Login. Upon Login There are 5 Tables to the left.



The login form is titled "ACT Analyzer Admin". It contains two input fields: "Username:" and "Password:". Below the password field is a blue "Log in" button.

Snap of the Various Tables that can be used to debug or understand completed/ongoing Transactions.

ACT Analyzer Admin		
Welcome to ACT Analyzer Portal		
ANALYZER_APIS		
Alert urls	+ Add	Change
Transaction alerts	+ Add	Change
Transaction datas	+ Add	Change
Transaction historys	+ Add	Change
Transaction paths	+ Add	Change
Transactions	+ Add	Change
AUTHENTICATION AND AUTHORIZATION		
Groups	+ Add	Change
Users	+ Add	Change

- 1. Transactions:** This table has details of all the Alerts that Analyze Module has received and the various status of each Alert/transaction.
 - **Transaction ID:** This is a unique ID Generated for each Transaction. This can be used to filter/search across all the tables.
 - **Trigger Node Name:** This is the Device against which the alert was received
 - **Trigger Interface Name:** This was the interface against with the alert was received
 - **Status:** This has the final status of the Transaction
 - **History:** This has all the historic status of the same transaction
 - **Transaction Issue:** This has The type of issue for which the Trigger was received (CHOK, CRC, FLAP).
 - **City:** This has the city details

Select Transaction to change

Q

Search

Action:

Go

0 of 2 selected

☐	TRANSACTION ID	TRIGGER NODE NAME	TRIGGER INTERFACE NAME	PATH FINDING NODE	PATH FINDING IP	REMOTE NODE NAME	REMOTE INTERFACE NAME	STATUS
☐	Bangalore_TV07nr1n	BLR-BEG-AX1	XGigabitEthernet0/0/24	<-Trigger Node	10.150.9.16	VBC-BX1	XGigabitEthernet0/0/21	REVERSE_REMEDIATION_INITIATED
☐	Bangalore_a5L1uPlp	BLR-BEG-AX1	XGigabitEthernet0/0/24	<-Trigger Node	10.150.9.16	VBC-BX1	XGigabitEthernet0/0/21	SOAK_REJECT

2 Transactions

Above is a snap of the Transactions Table. The Search Button on the top or Filter options on the right can be used to get to a specific Transaction.

2. Transaction Paths: This table has all the Alternate Paths the System has found for a Given transaction.

- **Transaction:** This is the Transaction ID Generated in the Transaction Table.
- **Node|Interface:** This is the Device|Interface Details of triggered Interface.
- **Path:** This gives the Node and Interface in the Path. For eg: (BLR-BEG-AX1 -> (XGigabitEthernet0/0/23) -> (XGigabitEthernet0/0/22) -> BLR-AKS-AX1 -> (Eth-Trunk11) -> (Eth-Trunk2) -> NOB-BX1)
- **Status:** Qualified/Rejected.
- **Created On:** Date when this was created
- **Problemnode|Probleminterface:** This has the Device and Interface details where the problem was seen in alternate Path and hence rejected.
- **Reason:** The reason for the Path being rejected is mentioned here. There are many reasons:
 - o Trigger link: This is the path where the issue is present, hence rejected
 - o Status Down: This is because an interface in the Path is down
 - o Optic Power out-of-range: This is because optic power is low
 - o CRC Errors: This is because one of the Links in alternate Path has CRC.

Search							
Action: ----- Go 0 of 2 selected							
IDN	NODE INTERFACE	PATH	COST	PROBLEMNODE PROBLEMINTERFACE	STATUS	CREATED ON	REASON
_TV07nr1n/BLR-BEG-AX1/XGigabitEthernet0/0/24/Bangalore	BLR-BEG-AX1 XGigabitEthernet0/0/24	BLR-BEG-AX1 -> (XGigabitEthernet0/0/23) -> (XGigabitEthernet0/0/22) -> BLR-AKS-AX1 -> (Eth-Trunk11) -> (Eth-Trunk2) -> NOB-BX1	0.0000		Qualified	April 14, 2022, 4:55 p.m.	-
_TV07nr1n/BLR-BEG-AX1/XGigabitEthernet0/0/24/Bangalore	BLR-BEG-AX1 XGigabitEthernet0/0/24	BLR-BEG-AX1 -> (XGigabitEthernet0/0/24) -> (XGigabitEthernet0/0/21) -> VBC-BX1	0.0000	BLR-BEG-AX1 XGigabitEthernet0/0/24	Rejected	April 14, 2022, 4:55 p.m.	Trigger Link

3. Transaction Datas: This table Gives the Details of the Operation performed on the Devices for a Specific Transaction.

- **Transaction ID:** This is the Transaction ID Generated in the Transaction Table.
- **Node|Interface:** This is the Device|Interface Details of triggered Interface

- **Cost Change Node Name:** This has the Node on which the Shutdown/Cost Change operation was triggered.
- **Cost Change Interface Name:** This has the Interface on which the Shutdown/Cost Change operation was triggered.
- **Operation:** This has the type of Operation. COST_CHANGE_REQUEST, SHUTDOWN_REQUEST
- **Status Flag:** Successful/unsuccessful
- **Reason Code:** reason for Failure
 - o Multiple VLAN Exists: This means that there was more than 1 VLAN passed in the p2p interface or bundle. Thus the tool could not identify the Correct Interface VLAN to increase Cost on
 - o Host Not reachable: Reachability issue

Bangalore_49oGB061 Search 3 results (3910 total)									
Action: Go 0 of 3 selected									
☐	TRANSACTION ID	NODEID INTERFACEID	COST CHANGE NODE NAME	COST CHANGE INTERFACE NAME	OPERATION	CONNECT TYPE	TIMESTAMP	STATUS FLAG	REASON
☐	Bangalore_49oGB061	BLR-HSR-AX8 XGigabitEthernet0/0/22	BLR-HSR-AX8	XGigabitEthernet0/0/22	cost_change	ssh	1650110906.4313107	unsuccessful	Multiple VLANs Exists
☐	Bangalore_49oGB061	BLR-HSR-AX8 XGigabitEthernet0/0/22	BLR-HSR-AX10	XGigabitEthernet0/0/23	COST_CHANGE_REQUEST			Successful	N/A
☐	Bangalore_49oGB061	BLR-HSR-AX8 XGigabitEthernet0/0/22	BLR-HSR-AX8	XGigabitEthernet0/0/22	COST_CHANGE_REQUEST			Successful	N/A

3 Transaction datas

- **Initial Cost:** This gives the Cost prior to increasing the cost
- **Final Cost:** This gives the Cost after increasing the Cost

Cost Change Details for a Transaction:

Bangalore_3GoSnBlu Search 4 results (3917 total)									
Action: Go 0 of 4 selected									
☐	TRANSACTION ID	NODEID INTERFACEID	COST CHANGE NODE NAME	COST CHANGE INTERFACE NAME	OPERATION	CONNECT TYPE	TIMESTAMP	STATUS FLAG	REASON CODE
☐	Bangalore_3GoSnBlu	ULL-BX1 XGigabitEthernet0/0/20	BLR-OID-AX1	XGigabitEthernet0/0/24	cost_change	ssh	1650112901.0558686	successful	NA
☐	Bangalore_3GoSnBlu	ULL-BX1 XGigabitEthernet0/0/20	ULL-BX1	XGigabitEthernet0/0/20	cost_change	ssh	1650112821.324233	successful	NA
☐	Bangalore_3GoSnBlu	ULL-BX1 XGigabitEthernet0/0/20	BLR-OID-AX1	XGigabitEthernet0/0/24	COST_CHANGE_REQUEST			Successful	N/A
☐	Bangalore_3GoSnBlu	ULL-BX1 XGigabitEthernet0/0/20	ULL-BX1	XGigabitEthernet0/0/20	COST_CHANGE_REQUEST			Successful	N/A

4 Transaction datas

Shutdown Details for a Transaction

☐	TRANSACTION ID	NODEID INTERFACEID	COST CHANGE NODE NAME	COST CHANGE INTERFACE NAME	OPERATION	CONNECT TYPE	TIMESTAMP	STATUS FLAG	REASON CODE
☐	Bangalore_Z7mLM10x	BLR-MKD-AX1 XGigabitEthernet0/0/24	KNP-BX1	XGigabitEthernet0/0/3	shutdown	ssh	1650112457.6061506	successful	NA
☐	Bangalore_Z7mLM10x	BLR-MKD-AX1 XGigabitEthernet0/0/24	KNP-BX1	XGigabitEthernet0/0/3	SHUTDOWN_REQUEST			Successful	N/A

2 Transaction datas

Work Chat Alerts

Alerts on Workchat are enabled for all the Transactions where remediation is performed with the status. Sample snap of the message below. Giving the details of the Transaction. The Transaction Id also is part of the message, so the teams can get more info on the Portal.

